

Aditya Khamitkar

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SUMMARY

Machine Learning Engineer & Data Scientist specializing in production LLM pipelines, real-time agentic orchestration, and high-performance systems. Demonstrated success in building low-latency AI architectures, executing RLHF evaluations, and developing systems-level C++ engines. Strong track record of translating complex exploratory research into highly optimized, deployed products.

EDUCATION

Vellore Institute of Technology (VIT) Sep 2022 – Jun 2027
Integrated M.Tech., Computer Science & Engineering ; Specialisation: Computational & Data Science

EXPERIENCE

AI Training Specialist & Data Science Consultant Jan 2025 – Present
Deccan AI Experts (Projects: MAITRIX, Shield, Bluebird, Action Extensions) *Remote*
– Spearheaded RLHF alignment pipelines for 4 frontier LLMs, designing structured evaluation rubrics that elevated Google Gemini instruction-following accuracy by **15%**.
– Diagnosed and eliminated **100%** of schema faults in an 18-tool agentic RL environment (Project MAITRIX) by authoring automated Python-driven SQLite audit harnesses.

AI Engineering Intern Oct 2025 – Dec 2025
Proton Engineering Consultancy & Services *Thane, India*
– Slashed HR screening time by **80%** (50 resumes in 42s vs. 120m) by architecting a real-time NLP service with SpaCy, pdfplumber, and Streamlit.
– Achieved **90%** candidate-categorisation accuracy through custom scoring functions and skill-clustering algorithms.

Data Science & ML Research Intern Oct 2024 – Nov 2024
Satish Dhawan Space Centre, ISRO *Sriharikota, India*
– Reduced manual signal-review overhead by **~40%** by deploying PyTorch/TensorFlow LSTM models against **4 PSLV/SSLV telemetry streams**.
– Engineered multi-source features to counter real-time atmospheric interference, enabling actionable degradation predictions.

OPEN SOURCE CONTRIBUTIONS

GirlScript Summer of Code (GSSoC) | Open Source Contributor 2024
– Actively contributed to external open-source repositories, resolving technical issues, optimizing codebase performance, and collaborating with global maintainers via GitHub pull requests.

PROJECTS & RESEARCH

Vocalis: Ultra-Low-Latency Voice Agent | Python, WebSockets, Deepgram STT, Groq API, asyncio 2026
– Architected a real-time streaming voice agent capable of handling user interruptions (barge-in) by managing asynchronous audio chunks over WebSockets.
– Achieved **500ms** end-to-end latency by optimizing the LLM inference pipeline with Groq and streaming transcriptions via Deepgram.

Synapse.cpp: MLP Engine from Scratch | C++17, RAII, Adam/SGD, Zero External Dependencies 2025
– Implemented a full forward/backward pass, Adam optimiser with bias correction, and gradient clipping entirely in raw C++17.
– Architected a RAII-compliant Matrix class with move semantics and flat storage, demonstrating production-grade memory management.

Nexus Scout: Multi-Agent Pipeline | LangGraph, LangChain, Pydantic, Streamlit 2025
– Designed a deterministic LangGraph multi-agent orchestration system with a weighted hybrid scoring engine to automate personalised outreach at scale.

BDH Ablations: Bio-Inspired Language Model | PyTorch, HuggingFace, WikiText-2 2026
– Isolated multiplicative gating as the dominant BDH component (**+0.059 nats loss** on ablation) through a rigorous 5-variant study of a dual-circuit SSM.

TECHNICAL SKILLS

Languages: Python (Proficient), C++17, SQL, Bash **CS Fundamentals:** OOP, LLD, Memory Management (RAII), Concurrency
Applied AI & Infrastructure: WebRTC, WebSockets, Streaming Audio, Deepgram, LangGraph, RLHF, RAG Pipelines, Multi-Agent Workflows
Machine Learning: PyTorch, TensorFlow, State-Space Models, LSTM, Time-Series Forecasting, Scikit-learn, XGBoost
Data Engineering & Tools: Apache Spark, PySpark, FastAPI, Docker, Kubernetes, AWS (EC2, S3), Git, MLflow

AWARDS & LEADERSHIP

Achievements: Intel AI4Youth National Finalist (Top 125 of 52k) Smart India Hackathon Finalist 2024 National Aerolympics 2020 Champion
Certifications: ML Engineering for Production (DeepLearning.AI) Deep Neural Networks with PyTorch (Coursera)
Leadership: Technical Team Head, Data Science Club VIT Active Member, ML Team & AI Club VIT